

- 3 (a) Define gravitational field at a point.

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 [1]

- (b) Fig. 3.1 shows an isolated point mass of mass M .

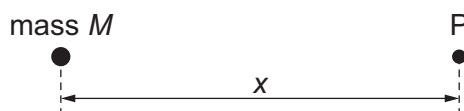


Fig. 3.1

Point P is at distance x from the point mass.

- (i) By considering the force exerted by the point mass on a test mass of mass m placed at P, derive an equation for the gravitational field strength g at P, in terms of M and x . Identify any other symbols you use.

[2]

- (ii) On Fig. 3.1, draw an arrow to indicate the direction of the gravitational field at P. [1]

- (iii) Point Q is at distance $\frac{x}{2}$ from the point mass, on the opposite side of the mass from P, as shown in Fig. 3.2.



Fig. 3.2

Compare the gravitational field at Q with that at P.

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 [2]

